Department of Mechanical Engineering

Course ME501: Thermodynamics and Fluid Mechanics

Credit Hours: 4 | L-T-P: 3-1-0 | Examination: End of Semester

Course Objective

To provide a working knowledge of the laws of thermodynamics and the principles governing the flow and application of fluids, and to enable students to apply them to the analysis of engineering systems.

Course Outcomes

Code	Course outcome	Bloom level
CO1	Apply the laws of thermodynamics to closed and open systems	Apply
CO2	Analyse the performance of power and refrigeration cycles	Analyse
CO3	Solve governing equations for pumping machinery	Apply
CO4	Evaluate power and	Evaluate

Unit 1: Thermodynamic Systems

Thermodynamic system, surroundings and boundary. Open and closed systems, isolated systems and adiabatic systems. State postulate. Zeroth law of thermodynamics and the concept of temperature.

Unit 2: Properties and Relations

Intensive and extensive properties. Specific heat at constant pressure and at constant volume. Characteristic gas equation. Availability and irreversibility.

Assessment: The assessment is 40 per cent internal and 60 per cent end of semester. Internal assessment is divided as follows:

Component	Marks	Weight
Mid-semester examination	20	20%
Assignment 1	6	6%
Assignment 2	6	6%
Laboratory record	8	8%

Unit 3: First and Second Laws

First law for a closed system, enthalpy and entropy. Second law, Carnot cycles, Clausius and Kelvin-Planck statements. Carnot efficiency for heat engines.

Unit 4: Fluid Mechanics

Continuity equation and Navier-Stokes equations in one dimension. Bernoulli equation and its applications. Friction losses in pipe flow, friction factor and Reynolds number.

Reforms: A student having attendance below 75 per cent is not eligible to appear in the end of semester examination in any subject.

Prerequisite: ME301 Engineering Thermodynamics. Students who have not completed the prerequisite will be de-registered at the start of the semester.